

Rickettsia parkeri Spotted Fever: A Comprehensive Disease Characterization

Disease: Rickettsia parkeri rickettsiosis (Rickettsia parkeri spotted fever) **MONDO ID:** MONDO:0000234 **Category:** Infectious Disease (tick-borne, spotted fever group rickettsiosis) **Report basis:** 5-iteration autonomous literature investigation; 14 confirmed findings; 34 papers reviewed

Summary

Rickettsia parkeri rickettsiosis is an emerging, tick-borne spotted fever group (SFG) rickettsiosis caused by the obligate intracellular Gram-negative bacterium *Rickettsia parkeri*. It presents as an acute, self-limited febrile illness whose clinical signature is an **inoculation eschar** (*tache noire*) at the tick-bite site, accompanied by a maculopapular—sometimes vesiculopustular—rash, regional lymphadenopathy, fever, headache, and myalgia. In a systematic review of 77 clinical cases, fever occurred in ~93%, eschar in ~87%, and rash in ~68% of patients ([PMID: 33989945](#)). The first US case series characterized the disease as "similar to but less severe than classically described Rocky Mountain spotted fever" (RMSF) ([PMID: 18808353](#)). Crucially, **no confirmed deaths have been reported worldwide** ([PMID: 42745336](#)), placing it at the benign end of the SFG rickettsiosis severity spectrum.

The disease is transmitted primarily by *Amblyomma maculatum* (the Gulf Coast tick) in the United States, and by *Amblyomma ovale* (the *R. parkeri* strain Atlantic rainforest), *Amblyomma triste*, and *Amblyomma tigrinum* in South America. Pathophysiology centers on **endothelial infection producing a localized necrotizing vasculitis** ("rickettsial vasculitis"), with the bacterium exploiting a distinctive **two-phase actin-based motility** system (RickA/Arp2/3 early; Sca2/formin-mimic late) to spread cell-to-cell. Host **type I interferon** is a key protective response; interferon-receptor-deficient mice are the tractable animal model that recapitulates the human eschar.

Because this is an infectious disease, the "genetic" sections of a heritable-disease template are **not applicable**: there is no human causal gene, no Mendelian inheritance, and no pathogenic human variant. The relevant genetics is that of the pathogen (e.g., *ompA*, *ompB*, *sca2*, *rickA*,

gltA). Diagnosis rests on eschar recognition plus IFA serology, PCR of the eschar crust, and immunohistochemistry; species-level confirmation requires molecular methods because SFG serology cross-reacts extensively. Treatment is **doxycycline**, to which cases respond uniformly. There is no vaccine, so prevention relies on tick-bite avoidance, prompt tick removal, and canine ectoparasite/vector control.

Key Findings

1. Clinical profile: an eschar-associated febrile illness milder than RMSF (F001)

The defining clinical portrait of *R. parkeri* rickettsiosis comes from a systematic review of 77 clinical cases (32 confirmed, 45 probable). Pooled phenotype frequencies were: **fever ~93%**, **eschar ~87%**, **rash ~68%**, **headache ~67%**, **myalgia ~61%**, with $\geq 60\%$ of cases showing altered laboratory parameters ([PMID: 33989945](#)):

"more than 60% of the cases had fever (mean of 93%), eschar (mean of 87%), and rash (mean of 68%). Headache and myalgia were predominant nonspecific symptoms (mean of 67% and 61%, respectively)"

The first 12 US cases (6 confirmed, 6 probable) established the disease as clinically distinct from — and milder than — RMSF ([PMID: 18808353](#)):

"The aggregate clinical characteristics of these patients revealed a disease similar to but less severe than classically described Rocky Mountain spotted fever."

It is the **second most prevalent SFG rickettsiosis** in the Americas.

Suggested HPO terms: Fever (HP:0001945), Skin ulcer/eschar (HP:0200042), Maculopapular exanthem/abnormality of the skin (HP:0000988), Lymphadenopathy (HP:0002716), Headache (HP:0002315), Myalgia (HP:0003326), Thrombocytopenia (HP:0001873), Elevated hepatic transaminases (HP:0002910), Lymphopenia (HP:0001888).

2. Vector biology and strain geography (F002)

R. parkeri exists as pathogenic strains with distinct vector–geography pairings:

Strain	Primary vector	Geography	Clinical note
<i>R. parkeri</i> sensu stricto	<i>Amblyomma maculatum</i> (Gulf Coast tick)	Southeastern/South-central USA	First human case 2004
Strain Atlantic rainforest	<i>Amblyomma ovale</i>	Brazil / South America	Milder, non-lethal, eschar
(regional)	<i>Amblyomma triste</i> , <i>A. tigrinum</i>	Argentina, Uruguay	Benign course

The Atlantic rainforest strain is explicitly characterized as causing "a milder non-lethal febrile disease with an eschar (necrosis) at the tick bite site" (PMID: 36693294). Vector range is **expanding northward**: *A. maculatum* + *R. parkeri* are now established in Ohio (PMID: 42661358) —

"We report the establishment of the Gulf Coast tick (*Amblyomma maculatum*) and *Rickettsia parkeri* bacteria in Ohio, USA."

— and in southern Illinois, where *R. parkeri* infection prevalence in *A. maculatum* reached 16.4% (PMID: 42314659). Laboratory competence has also been shown for *A. americanum* and *Dermacentor variabilis* (PMID: 41749360). Dogs serve as sentinel hosts.

3. Core pathophysiology: endothelial infection and two-phase actin-based motility (F003)

R. parkeri is an obligate intracellular Gram-negative bacterium with tropism for **vascular endothelium**. Infection produces vascular inflammation, loss of vascular integrity, and increased permeability — collectively "rickettsial vasculitis" (PMID: 19327117):

"the pathogen's affinity for endothelium lining the blood vessels, the consequences of which are vascular inflammation, insult to vascular integrity and compromised vascular permeability, collectively termed 'Rickettsial vasculitis'"

Intracellularly, *R. parkeri* escapes the phagosome into the cytoplasm and drives **actin-based motility in two temporally distinct phases** (PMID: 24361066):

"each protein directs an independent mode of *Rickettsia parkeri* motility at different times during infection. Early after invasion, motility is slow and meandering, generating short, curved actin tails that are enriched with Arp2/3 complex and cofilin"

Early motility is driven by **RickA** activating the host Arp2/3 complex; late, fast, directionally persistent motility requires **Sca2**, a functional mimic of eukaryotic formins (PMID: 20972427):

"Sca2 nucleates unbranched actin filaments, processively associates with growing barbed ends, requires profilin for efficient elongation, and inhibits the activity of capping protein, all properties shared with formins"

Structural work (cryo-EM) shows Sca2 forms a formin-FH2-like doughnut core (PMID: 37028467), and comparative studies reveal divergent motility mechanisms across *Rickettsia* species (e.g., *R. bellii* Sca2/6; PMID: 42048062).

Suggested GO/CL terms: actin nucleation (GO:0045010), Arp2/3 complex-mediated actin nucleation (GO:0034314), actin filament-based movement (GO:0030048), host cell cytoplasm (GO:0030430), endothelial cell (CL:0000115).

4. Immune mechanism: type I interferon is protective (F004)

Innate immunity controls *R. parkeri* infection. **Interferon α/β receptor-deficient mice** (and combined IFN-receptor-deficient mice) are susceptible and develop eschar-associated rickettsiosis, whereas immunocompetent wild-type mice are largely resistant (PMID: 34423779):

"Interferon receptor-deficient mice are susceptible to eschar-associated rickettsiosis"

The bacterium in turn **antagonizes type I interferon via inflammasome activity**, and this antagonism enhances pathogenesis (PMID: 32123346):

"Inflammasome-mediated antagonism of type I interferon enhances *Rickettsia* pathogenesis"

This host–pathogen cross-talk (interferons + inflammasomes) determines the balance between control and disease.

Suggested GO terms: type I interferon signaling pathway (GO:0060337), response to interferon-alpha (GO:0035455), inflammasome complex (GO:0061702).

5. Diagnosis and treatment (F005)

Doxycycline is the treatment of choice for SFG rickettsioses in both adults and children, and early empiric therapy prevents severe morbidity and death ([PMID: 16572105](#)):

"the recommendations for doxycycline are the treatment of choice for both adults and children"

R. parkeri cases respond uniformly and promptly. **Diagnosis** relies on: (1) recognition of the inoculation eschar; (2) indirect immunofluorescence antibody assay (IFA) demonstrating seroconversion / ≥ 4 -fold titer rise; (3) PCR of eschar crust or skin biopsy targeting *ompA*, *gltA*, *ompB*; (4) immunohistochemistry; and (5) cell-culture isolation ([PMID: 18808353](#)):

"Using indirect immunofluorescence antibody assays, immunohistochemistry, polymerase chain reaction assays, and cell culture isolation, we identified 6 confirmed and 6 probable cases of infection with *R. parkeri*."

Because SFG rickettsiae share extensive antigenic cross-reactivity, **species-specific confirmation requires PCR/sequencing** (or cross-adsorption serology) ([PMID: 20404224](#)):

"The SFG rickettsioses share many clinical manifestations and extensive antigenic cross-reactivity that may hamper specific confirmation of the causative agent."

Common laboratory abnormalities: mild thrombocytopenia, lymphopenia, and elevated hepatic transaminases ($\geq 60\%$ of cases have altered labs; [PMID: 33989945](#)).

Suggested NCIT terms: Doxycycline (NCIT:C305); Antibiotic Therapy (NCIT:C15844); Polymerase Chain Reaction (NCIT:C17003); Immunohistochemistry (NCIT:C16681).

6. Epidemiology (F006)

R. parkeri rickettsiosis is a reportable SFG rickettsiosis primarily affecting **adults, especially males aged 18–64**, with tick-exposure history in >90% of cases; confirmed/probable cases have been reported in the United States, Argentina, Brazil, Uruguay, and Colombia ([PMID: 33989945](#)):

"our results show that *R. parkeri* rickettsiosis is more frequent in males in the age group of 18-64 years and that a history of tick exposure was frequent (>90%). Cases were described in the United States, Argentina, Brazil, Uruguay and Colombia"

In the US it is reported under the nationally notifiable aggregate category "spotted fever rickettsiosis" (species-nonspecific due to serologic cross-reactivity). US tickborne bacterial/protozoan disease reports more than doubled from 2004 to 2016 ([PMID: 29723166](#)). In Argentina, two epidemiologic scenarios coexist — severe *R. rickettsii* (*Amblyomma cajennense* complex) versus **benign *R. parkeri*** (vectors *A. triste*, *A. tigrinum*) ([PMID: 30928146](#)):

"*R. parkeri* produces benign and self-limited clinical manifestation"

7. Temporal course and prognosis (F007)

Onset is **acute, typically 7–10 days after the tick bite**. Patients develop a progressively enlarging inoculation eschar, then exanthem/maculopapular (sometimes vesiculopustular) rash, regional lymphadenopathy, fever, myalgia, and prostration ([PMID: 42745336](#)):

"All patients developed symptoms 7-10 days after the bite, including inoculation eschar with progressive enlargement, exanthema, regional lymphadenopathy, fever, myalgia, and prostration; all responded to doxycycline treatment"

The disease is **self-limited and benign**, and across all reported cases worldwide **none have been fatal**:

"So far, none of the cases have been associated with death"

This contrasts sharply with RMSF (*R. rickettsii*), which carries a high case-fatality rate if untreated (pediatric under-treatment concerns highlighted in [PMID: 24252781](#)).

8. Prevention (F008)

No licensed human vaccine exists for any SFG rickettsiosis; prevention is behavioral and environmental. Primary prevention includes avoiding tick habitat, using repellents (DEET, permethrin-treated clothing), performing tick checks, and prompt removal of attached ticks. Because dogs are hosts/sentinels that bring vector ticks into peridomestic settings, control programs emphasize canine ectoparasite control and public awareness ([PMID: 35293560](#)):

"the results of the present study indicate the importance of implementing programs to control canine ectoparasites and to raise awareness of the risks of infection, signs and symptoms of SF caused by *R. parkeri* strain Atlantic Rainforest"

Secondary prevention is early eschar recognition and prompt empiric doxycycline. Antibiotic prophylaxis after tick bite is **not** recommended for SFG rickettsioses ([PMID: 16572105](#)).

9. Taxonomy and zoonotic ecology (F009)

Causative agent: *Rickettsia parkeri* Lackman et al., 1965 (NCBI Taxonomy ID 35792), an obligate intracellular Gram-negative alphaproteobacterium of the spotted fever group. It is a tick-borne **zoonosis with no human-to-human transmission**. Ticks are both vector and reservoir, maintaining the bacterium transstadially and transovarially. Dogs are excellent sentinels ([PMID: 42019180](#)):

"Dogs have been recognized as good sentinels for these rickettsioses, since they are hosts to different species and stages of ticks, and they sustain a good immunological response after infection"

Natural infection is documented across diverse tick species (*A. maculatum*, *A. ovale*, *A. triste*, *A. tigrinum*, *A. nodosum* strain NOD, *A. parvum*) and in wild carnivores/small mammals across the Americas ([PMID: 41133807](#); [PMID: 41880873](#)):

"three adults of *A. nodosum* contained DNA of *Rickettsia parkeri* Lackman et al., 1965 strain NOD"

10. Anatomical structures affected (F010)

The **primary anatomical lesion is the skin at the tick-bite site** — an inoculation eschar (*tache noire*): a central zone of dermal–epidermal necrosis with surrounding perivascular lymphohistiocytic inflammation and small-vessel (necrotizing) vasculitis, the histopathologic hallmark of SFG rickettsioses (PMID: 20404224):

"We report 3 cases of SFG rickettsiosis and discuss the epidemiology, clinical presentation, histopathologic features, and laboratory findings that support confirmed or probable diagnoses of *R parkeri* infection"

Regional (draining) **lymph nodes** are frequently enlarged, and the disseminated rash reflects widespread small-vessel endothelial infection. The **cellular target is the vascular endothelial cell (CL:0000115)**; the bacterium replicates free in the host-cell cytoplasm (GO:0005737). Unlike RMSF, systemic organ involvement (lung, brain, kidney) is minimal or absent because disease remains largely localized and self-limited.

Suggested UBERON terms: skin of body (UBERON:0002097), dermis (UBERON:0002067), endothelium of blood vessel (UBERON:0001986), lymph node (UBERON:0000029), integumentary system (UBERON:0002416).

11. Genetic/molecular basis is bacterial, not human (F011)

R. parkeri rickettsiosis is an infectious disease with **no human germline/somatic causal gene, no Mendelian inheritance, no pathogenic human variant, and no chromosomal abnormality** — the genetic sections of a heritable-disease template are **not applicable**. Host genetics contributes only to susceptibility via innate immune pathways (e.g., type I interferon signaling; PMID: 34423779). The relevant genetics is that of the **pathogen**:

Gene	Product / role
<i>ompA</i> (sca0)	Outer membrane protein A; adhesin; diagnostic/typing target
<i>ompB</i> (sca5)	Outer membrane protein B; cell entry
<i>sca2</i>	Formin-mimic driving late actin-based motility (PMID: 20972427)
<i>rickA</i>	Arp2/3 activator driving early motility (PMID: 24361066)
<i>gltA</i>	Citrate synthase; molecular typing
<i>sca4</i> (gene D)	Surface cell antigen

Phylogenetics supports multiple distinct *R. parkeri* strains in the New World ([PMID: 29439989](#)), and inter-tick-species sequence differences occur in cell-entry genes ([PMID: 41749360](#)):

"Differences in cellular-entry and pathogen chromosomal genes (OmpA, OmpB, and 16S) were detected within the different tick species"

12. Identifiers and synonyms (F012)

Preferred name: *Rickettsia parkeri* rickettsiosis (MONDO:0000234).

Identifier	Value
MONDO	MONDO:0000234
ICD-10	A77.8 "Other spotted fevers"
ICD-11	1C30.2 "Spotted fever due to other <i>Rickettsia</i> species" (or 1C30.Y)
MeSH	No species-specific descriptor; indexed under "Rickettsia Infections" (D012373); organism " <i>Rickettsia parkeri</i> "
OMIM	None (not a heritable disease)
Orphanet	No dedicated number (not a rare genetic disorder)
NCBI Taxonomy (agent)	35792 (<i>Rickettsia parkeri</i>)

Synonyms: *R. parkeri* spotted fever; Tidewater spotted fever; American boutonneuse fever; "maculatum infection/disease"; and for the South American strain, Atlantic rainforest spotted fever / mata atlântica spotted fever ([PMID: 18808353](#); [PMID: 36693294](#)):

"*Rickettsia parkeri* rickettsiosis, a recently identified spotted fever transmitted by the Gulf Coast tick (*Amblyomma maculatum*), was first described in 2004."

Information source: aggregated disease-level literature (case series, systematic reviews, surveillance) — not single-patient EHR resources.

13. Model systems (F013)

- **In vivo:** Type I/II interferon-receptor-deficient mice (*Ifnar1*^{-/-}, *Ifnar1*^{-/-}*Ifngr1*^{-/-}) are susceptible and develop eschar-associated rickettsiosis after intradermal inoculation — the first tractable mammalian model recapitulating the human eschar ([PMID: 34423779](#)). Immunocompetent wild-type mice are largely resistant (a key limitation).
- **In vitro:** Vero cells and human microvascular endothelial cells allow direct visualization of actin-based motility and cell-to-cell spread ([PMID: 24361066](#); [PMID: 20972427](#)); tick cell lines model the arthropod host.
- **Historical:** Guinea pigs model SFG rickettsial fever/scrotal reaction.

14. Consolidated conclusion (F014)

Across all evidence, *R. parkeri* rickettsiosis is a mild, self-limited, tick-borne SFG rickettsiosis (fever ~93%, eschar ~87%, rash ~68%; [PMID: 33989945](#)), with no confirmed deaths ([PMID: 42745336](#)) and uniform doxycycline responsiveness ([PMID: 16572105](#)). The single most useful diagnostic discriminator from RMSF is the **inoculation eschar**, characteristic of *R. parkeri* and essentially absent in RMSF; the first US series framed the disease by "its clinical distinction from Rocky Mountain spotted fever" and found it "similar to but less severe than" RMSF ([PMID: 18808353](#)).

Mechanistic Model / Interpretation

Ordered causal chain (initiating exposure → clinical manifestation)

1. Infected Amblyomma tick attaches and feeds on human skin (7–10 day incubation)
 - | leads to
2. *R. parkeri* inoculated into the dermis at the bite site
 - | results in
3. Bacteria adhere to and invade vascular ENDOTHELIAL cells
 - | (adhesins OmpA/OmpB; cell entry) – results in
4. Phagosomal escape → replication free in host-cell CYTOPLASM
 - | leads to
5. Two-phase ACTIN-BASED MOTILITY:
 - ├ EARLY: RickA activates host Arp2/3 → slow, meandering motility
 - └ LATE: Sca2 (formin-mimic) → fast, directional cell-to-cell spread
 - | results in
6. Local endothelial infection + host innate immune response
 - | (type I IFN protective; bacterium antagonizes IFN via inflammasome)
 - | leads to
7. Focal NECROTIZING VASCULITIS of small dermal vessels
 - | results in
8. INOCULATION ESCHAR (tache noire) at bite site ← hallmark lesion
 - | and (branch) limited hematogenous/lymphatic spread
 - ├ leads to → maculopapular/vesiculopustular RASH (disseminated endothelium)
 - ├ leads to → regional LYMPHADENOPATHY (draining nodes)
 - └ leads to → systemic FEVER, headache, myalgia; mild lab abnormalities
 - | In immunocompetent hosts:
9. Host immune control (+ doxycycline) → SELF-LIMITED RESOLUTION, no death

Upstream vs downstream: Steps 1–5 (inoculation, endothelial invasion, cytoplasmic replication, actin motility) are upstream drivers; steps 7–8 (vasculitis, eschar) are the proximate downstream lesions producing the clinical phenotype. The interferon axis (step 6) is a modulating branch that determines whether infection is contained (immunocompetent hosts, most humans) or progresses (interferon-receptor-deficient mice). Steps 3–5 are directly demonstrated in cell culture; step 6's protective role is demonstrated in mouse knockouts and *inferred* for humans.

Why *R. parkeri* is milder than RMSF (interpretation): The disease remains largely **localized** to the inoculation site and skin, with minimal systemic endothelial injury to lung/brain/kidney — the opposite of the widespread, high-permeability endothelial damage that makes *R. rickettsii* (RMSF) lethal. The eschar is therefore both the pathologic signature and a marker of contained, localized disease.

Comparison table: *R. parkeri* vs RMSF (*R. rickettsii*)

Feature	<i>R. parkeri</i> rickettsiosis	Rocky Mountain spotted fever
Vector	<i>A. maculatum</i> , <i>A. ovale</i> , <i>A. triste</i>	<i>Dermacentor</i> spp., <i>A. sculptum</i>
Inoculation eschar	Present (~87%) — hallmark	Typically absent
Rash	Maculopapular/vesiculopustular	Maculopapular → petechial
Systemic organ injury	Minimal / localized	Severe, multi-organ
Case fatality	~0% (no confirmed deaths)	High if untreated
Treatment	Doxycycline (uniform response)	Doxycycline (urgent)

Evidence Base

PMID	Title (abbrev.)	Supports
33989945	<i>Clinical/epidemiological/lab features: systematic review</i>	Pooled phenotype frequencies; demographics (F001, F006)
18808353	<i>R. parkeri rickettsiosis and its clinical distinction from RMSF</i>	Milder-than-RMSF; diagnostics; name/first description (F001, F005, F012, F014)
36693294	<i>Inoculation eschar in Brazil</i>	Atlantic rainforest strain/vector; eschar (F002, F012)
42661358	<i>Gulf Coast ticks + R. parkeri, Ohio</i>	Range expansion (F002)
42314659	<i>Surveillance, southern Illinois</i>	16.4% tick infection; range expansion (F002, F006)
19327117	<i>Host-cell interactions with pathogenic Rickettsia</i>	Endothelial tropism; rickettsial vasculitis (F003, F010)
24361066	<i>Actin motility in distinct phases</i>	Two-phase RickA/Sca2 motility (F003, F011, F013)
20972427	<i>Sca2 is a bacterial formin-like mediator</i>	Sca2 formin-mimic mechanism (F003, F011, F013)
37028467	<i>Cryo-EM of Sca2 formin-like core</i>	Sca2 structure (F003)
42048062	<i>Divergent Rickettsia motility mechanisms</i>	Species divergence in motility (F003)
34423779	<i>IFN receptor-deficient mice susceptible</i>	Protective type I IFN; eschar mouse model (F004, F013)
32123346	<i>Inflammasome antagonism of type I IFN</i>	Immune cross-talk enhances pathogenesis (F004)
16572105	<i>Diagnosis/management of tickborne rickettsial diseases</i>	Doxycycline first-line; no prophylaxis (F005, F008)
20404224	<i>Expanding spectrum of eschar-associated rickettsioses</i>	Histopathology; serologic cross-reactivity (F005, F010)
30928146	<i>Spotted fever in Argentina</i>	Benign course; Argentine vectors (F006, F007)
42745336	<i>First outbreak of Atlantic Rainforest SF in NE Brazil</i>	Incubation, symptom sequence; no deaths (F007, F014)

PMID	Title (abbrev.)	Supports
35293560	<i>New focus of SF by R. parkeri in Brazil</i>	Canine ectoparasite control / prevention (F008)
42019180	<i>SFG rickettsiae in dogs of the Americas: meta-analysis</i>	Dogs as sentinels; zoonotic ecology (F009)
41133807	<i>Rickettsia in horses/ticks, Pernambuco</i>	Multi-tick natural infection; species authority (F009)
41880873	<i>Rickettsia in Cerrado carnivores</i>	Wild reservoir ecology (F009)
29439989	<i>Multiple strains of R. parkeri in the New World</i>	Pathogen strain diversity (F011)
41749360	<i>R. parkeri diversity from three tick species</i>	Cell-entry gene variation; vector competence (F002, F011)
29723166	<i>Vital Signs: vectorborne disease trends 2004–2016</i>	Rising tickborne disease burden (F006)
23440128	<i>R. parkeri in A. triste, Argentina</i>	Molecular typing targets; Argentine vector (F011)
24252781	<i>Provider treatment practices, RMSF</i>	Context: RMSF lethality/doxycycline under-use

Evidence source types: Human clinical (case series, systematic reviews: 33989945, 18808353, 42745336, 30928146, 20404224); epidemiologic/surveillance (42314659, 42661358, 29723166); veterinary/ecologic (42019180, 41880873, 41133807, 32267390); in vitro / cell biology (24361066, 20972427, 37028467, 42048062); model organism (34423779, 32123346); phylogenetic/computational (29439989, 41749360, 23440128).

Limitations and Knowledge Gaps

- 1. Under-recognition and surveillance dilution.** In the US, *R. parkeri* cases are folded into the aggregate "spotted fever rickettsiosis" notifiable category because SFG serology cross-reacts; true incidence and prevalence (cases per 100,000) are therefore **not precisely quantified**. Species-level burden is likely underestimated.

2. **Small evidence base.** The definitive clinical picture rests on 77 pooled cases ([PMID: 33989945](#)) plus scattered series. No large prospective cohorts or randomized treatment trials exist.
 3. **No confirmed deaths, but rare severe cases possible.** The "0% fatality" statement reflects reported cases; immunocompromised or atypical presentations may be under-captured.
 4. **Model limitations.** The tractable animal model requires interferon-receptor knockout mice; immunocompetent mice resist infection, so the model does not fully reproduce the mild human course or long-term immunity.
 5. **Quality-of-life / long-term outcome data are absent.** Because disease is self-limited, no EQ-5D/SF-36/PROMIS data or disability outcomes have been reported.
 6. **Human genetic susceptibility is essentially unstudied.** Beyond the interferon axis inferred from mouse work, no human host-genetic risk/protective factors, GWAS loci, or pharmacogenomic data exist for this disease.
 7. **Strain–virulence correlations incomplete.** Whether sequence differences in *ompA/ompB* across strains/tick species translate into clinical severity differences is not established.
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Proposed Follow-up Experiments / Actions

1. **Species-specific surveillance:** Advocate for molecular (PCR/sequencing) confirmation in reportable SFG cases to disaggregate *R. parkeri* from other SFG agents and generate true incidence/prevalence estimates by region.
2. **Prospective clinical registry:** Establish a multi-country (US + South America) registry capturing incubation, symptom frequencies, lab abnormalities, treatment response, and any severe/atypical outcomes to refine natural-history and prognosis data.
3. **Strain-virulence genomics:** Comparative genomics/transcriptomics of *R. parkeri* sensu stricto vs Atlantic rainforest vs NOD strains, correlating *ompA/ompB/sca2/rickA* variation with in vitro endothelial cytopathology and mouse eschar severity.
4. **Host immunity mechanism:** Dissect the type I interferon ↔ inflammasome axis (building on [PMID: 34423779](#), [PMID: 32123346](#)) to explain why *R. parkeri* is contained (eschar-limited) whereas *R. rickettsii* disseminates.
5. **Vector-range monitoring:** Continue active tick surveillance in newly colonized northern areas (Ohio, Illinois) to forecast human risk as *A. maculatum* expands ([PMID: 42661358](#), [PMID: 42314659](#)).

6. **One-Health prevention trials:** Evaluate canine ectoparasite-control programs as a community-level intervention to reduce peridomestic vector burden ([PMID: 35293560](#)).
7. **Diagnostic assay development:** Develop rapid, field-deployable species-specific molecular assays for eschar-crust testing to overcome serologic cross-reactivity ([PMID: 20404224](#)).

Ontology Annotation Summary

Category	Terms
Disease	MONDO:0000234 (<i>Rickettsia parkeri</i> rickettsiosis)
Phenotypes (HPO)	HP:0001945 (Fever), HP:0200042 (Skin ulcer/eschar), HP:0002716 (Lymphadenopathy), HP:0002315 (Headache), HP:0003326 (Myalgia), HP:0001873 (Thrombocytopenia), HP:0001888 (Lymphopenia), HP:0002910 (Elevated hepatic transaminases)
Cell types (CL)	CL:0000115 (endothelial cell)
Anatomy (UBERON)	UBERON:0002097 (skin), UBERON:0002067 (dermis), UBERON:0001986 (endothelium of blood vessel), UBERON:0000029 (lymph node)
Biological process (GO)	GO:0030048 (actin filament-based movement), GO:0034314 (Arp2/3-mediated actin nucleation), GO:0045010 (actin nucleation), GO:0060337 (type I interferon signaling), GO:0061702 (inflammasome complex)
Cellular component (GO)	GO:0005737 / GO:0030430 (cytoplasm / host cell cytoplasm)
Chemical (CHEBI)	CHEBI:50845 (doxycycline)
Treatment (NCIT)	NCIT:C305 (Doxycycline), NCIT:C15844 (Antibiotic Therapy)
Pathogen (NCBI Taxon)	35792 (<i>Rickettsia parkeri</i>)

Report compiled from 14 confirmed findings and 34 reviewed papers across a 5-iteration autonomous investigation. All quoted material is verbatim from the cited PubMed abstracts.

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